National University

of Computer & Emerging Sciences-Faisalabad

Syllabus: ⊙ Flowcharts ⊙ Pseudocodes

CS1002-Programming Fundamentals

Quiz_01

Total Time: 15 min **Total Marks: 20** 13

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Q1. CLO1 **10 + 10**

a. Write the pseudocode for a program that reads in the radius of a circle from the user after displaying proper prompt. Then it asks if the user wants to print the circle's diameter (1), circumference (2) or area (3), or all the above (4). Use the constant value 3.14 for π. This task will be repeated **N** times. N will be input by the user as well. Finally print a message of "Goodbye" to the use.

Note: Take input of 1, 2, 3 or 4 for the options given to the user.

Hint: Use nested IF-ELSE and WHILE/ENDWHILE.

1.0 Start

2.0 Declare r, count, dia, area, circumference

3.0 Print" Enter ~~radius of circle and enter~~ counter value"

4.0 Input count

5.0 While count > 0 then

5.1 Print" Enter radius of circle"

5.2 Input radius and save in r

5.3 If user wants the print of:

(1) diameter

(2) circumference

(3) Area

(4) Both

5.3.1 Input option

5.3.2 If (1) diameter then

5.3.2.1 dia ← 2*r

5.3.2.2 Print" Dia" dia

5.3.3 Else ~~then~~ if

5.3.3.1 if (2) then

5.3.3.1.1 circumference ← 2(3.14)(r)

5.3.3.1.2 Print "circumference" circumference

5.3.3.2 Else ~~then~~

5.3.3.2.1 if (3) then

5.3.3.2.1.1 Area ← (3.14)*(r)²

5.3.3.2.1.2 Print "A" Area

5.3.3.2.2 Else then

5.3.3.2.1 if (4)

5.3.3.2.2 dia ← 2r

Area ← (3.14)*r²

Circum ← 2(3.14)r.

5.3.3.2.2.1 Print" Area dia

5.4 Endif circum

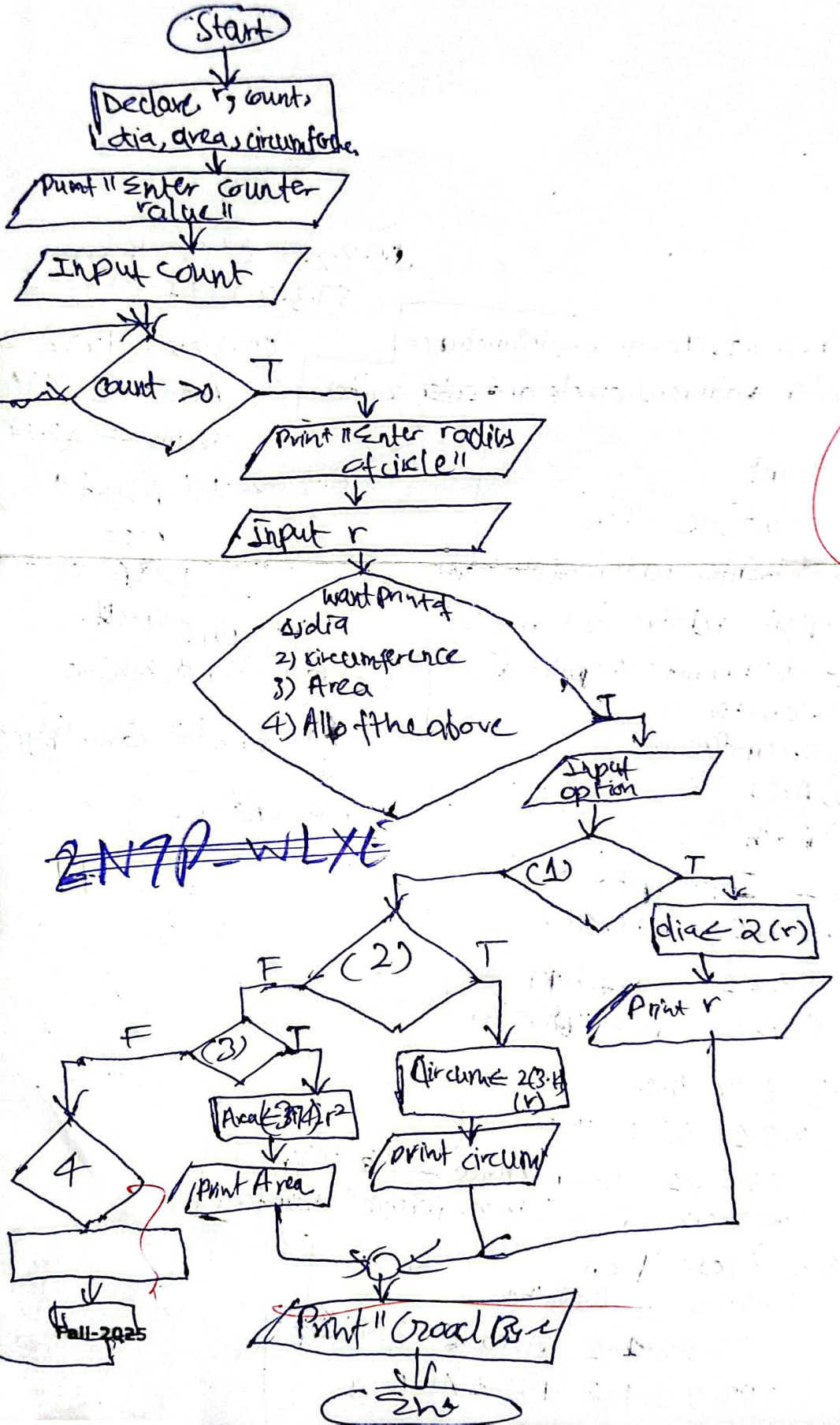
6.0 End While

7.0 Print" Goodbye"

8.0 End.

6+1

b. Now draw the flow chart of the above written pseudocode. The flowchart will be checked on the criterion that it matches the pseudocode or not.



6